

Column/Row operations

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Learning Objectives

Matrix Multiplication(1.4)



Row/Column Operations(1.3)

- What **three** kind of row/column operations are there?
- For the product $AB = C$,
 - Which two matrices is allowed to play **simultaneous row operation** without changing the equality?
 - Which two matrices is allowed to play **simultaneous column operation** without changing the equality?
- Will simultaneous (row/column) operation changes $A^{-1}B$ or AB^{-1} ?
- Let E be a matrix obtained by applying some row operations from I , what does EA mean?
- Let E be a matrix obtained by applying some column operations from I , what does AE mean?
- Why matrix equation $Ax = b$ is a system of linear equations?
- How row operation help with solving system of linear equations?
 - What kind of matrix are we reducing to?
- How to use row/column operations to find **inverse**?

A special request

A customer requests for a special **new drink** need the following ingredients

Old Drinks ingredients:

				
	0	2	0	2
	0	1	0	1
	0	4	2	0
	1	1	0	0

New Drink requirement:

	
	4
	2
	2
	4

Problem:

	
	?
	?
	?
	?

But the chef only have  ,  ,  ,  at the hand, can he produce  by those materials?

The chef thought this problem is the same as a matrix product equation, indeed, replace those questionmarks by x, y, z , he need the following equation to be true

Old Drinks ingredients:

				
	0	2	0	2
	0	1	0	1
	0	4	2	0
	1	1	0	0

$\times$

Problem:

	
	x
	y
	z
	w

$=$

New Drink requirement:

	
	4
	2
	2
	4

Mathematically, this equation is writing as

$$\begin{pmatrix} 0 & 2 & 0 & 2 \\ 0 & 1 & 0 & 1 \\ 0 & 4 & 2 & 0 \\ 1 & 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \\ w \end{pmatrix} = \begin{pmatrix} 4 \\ 2 \\ 2 \\ 4 \end{pmatrix}$$

By understanding by columns, solving it is the same as asking for

$$\begin{array}{ccccccccc} 4 & & 0 & & 2 & & 0 & & 2 \\ 2 & & 0 & & 1 & & 0 & & 1 \\ 2 & = & 0 & \mathbf{x} + & 4 & \mathbf{y} + & 2 & \mathbf{z} + & 0 & \mathbf{w} \\ 4 & & 1 & & 1 & & 0 & & 0 \end{array}$$

This can be write as the following and we call it the **Linear equation**

$$\left\{ \begin{array}{l} 0x + 2y + 0z + 2w = 4 \\ 0x + 1y + 0z + 1w = 2 \\ 0x + 4y + 2z + 0w = 2 \\ 1x + 1y + 0z + 0w = 4 \end{array} \right.$$

Changing materials

Observation: The question is only asking for the meal's demand for semi-product meals. It does not asking anything related to the raw material.

	
	x
	y
	z
	w

No  ,  ,  ,  appeared in this question. Therefore we can change materials to **simplify** the problem.

Row reduction

The clever chef changes the **material** so the ingredients table is easier

					
	0	2	0	2	4
	0	1	0	1	2
	0	4	2	0	2
	1	1	0	0	4

					
	1	1	0	0	4
	0	2	1	0	1
 + 	0	1	0	1	2
	0	0	0	0	0

$$\begin{cases} 0x + 2y + 0z + 2w = 4 \\ 0x + 1y + 0z + 1w = 2 \\ 0x + 4y + 2z + 0w = 2 \\ 1x + 1y + 0z + 0w = 4 \end{cases}$$

$$\Rightarrow \begin{cases} 1x + 1y + 0z + 0w = 4 \\ 0x + 2y + 1z + 0w = 1 \\ 0x + 1y + 0z + 1w = 2 \\ 0x + 0y + 0z + 0w = 0 \end{cases}$$

Row Multiplying

Doubling the material  $\mapsto$  will multiply 3rd row by $\frac{1}{2}$.

					
	0	2	0	2	4
	0	1	0	1	2
	0	4	2	0	2
	1	1	0	0	4

					
	0	2	0	2	4
	0	1	0	1	2
	0	2	1	0	1
	1	1	0	0	4

$$\left\{ \begin{array}{l} 0x + 2y + 0z + 2w = 4 \\ 0x + 1y + 0z + 1w = 2 \\ 0x + 4y + 2z + 0w = 2 \\ 1x + 1y + 0z + 0w = 4 \end{array} \right. \Rightarrow \left\{ \begin{array}{l} 0x + 2y + 0z + 2w = 4 \\ 0x + 1y + 0z + 1w = 2 \\ 0x + 2y + 1z + 0w = 1 \\ 1x + 1y + 0z + 0w = 4 \end{array} \right.$$

On equation, the third equation has been divided by 2. This is called **row multiplying**

Row Switching

Switching the order would not change problem,

					
	0	2	0	2	4
	0	1	0	1	2
	0	2	1	0	1
	1	1	0	0	4

					
	1	1	0	0	4
	0	2	1	0	1
	0	1	0	1	2
	0	2	0	2	4

$$\begin{cases} 0x + 2y + 0z + 2w = 4 \\ 0x + 1y + 0z + 1w = 2 \\ 0x + 2y + 1z + 0w = 1 \\ 1x + 1y + 0z + 0w = 4 \end{cases} \Rightarrow \begin{cases} 1x + 1y + 0z + 0w = 4 \\ 0x + 2y + 1z + 0w = 1 \\ 0x + 1y + 0z + 1w = 2 \\ 0x + 2y + 0z + 2w = 4 \end{cases}$$

This is called **row switching**

Row Adding

Replacing 🍊 by 🍊🍃🍃. Each time I use the package 🍊🍃🍃 I save 🍃🍃. Then the row for 🍃 is reduced by 2 times the row for 🍊.

	🥛	🍲	☕	🥣	🍷
🐮	1	1	0	0	4
🍫 🍫	0	2	1	0	1
🍊	0	1	0	1	2
🍃	0	2	0	2	4

	🥛	🍲	☕	🥣	🍷
🐮	1	1	0	0	4
🍫 🍫	0	2	1	0	1
🍊🍃🍃	0	1	0	1	2
🍃	0	0	0	0	0

$$\begin{cases} 1x + 1y + 0z + 0w = 4 \\ 0x + 2y + 1z + 0w = 1 \\ 0x + 1y + 0z + 1w = 2 \\ 0x + 2y + 0z + 2w = 4 \end{cases} \Rightarrow \begin{cases} 1x + 1y + 0z + 0w = 4 \\ 0x + 2y + 1z + 0w = 1 \\ 0x + 1y + 0z + 1w = 2 \\ \mathbf{0x + 0y + 0z + 0w = 0} \end{cases}$$

On equation, 4'th equation has been subtracted by 3rd equation. This is called **row adding**

When should row reduction stop?

In one words, row operation is **updating the raw ingradient list**.

We should stop if the matrix is **simple enough** for us to solve equations.
So what is simple? There are many discussions.

Theory 1: We should stop if

- Each non-zero row has an entry 1, such that this 1 is the only non-zero entry on its columns.

					
	1	1	0	0	4
	0	2	1	0	1
	0	1	0	1	2
	0	0	0	0	0

Explanation to Theory 1: If the matrix has been reduced that way, we obtain an equation

$$\begin{cases} 1x + 1y + 0z + 0w = 4 \\ 0x + 2y + 1z + 0w = 1 \\ 0x + 1y + 0z + 1w = 2 \end{cases}$$

such that each equation has a variable, such that this variable does not appear in other equations. When this happens, we may move these variable to one side of equation. Assigning other variables arbitrary values would end up with a solution.

$$\begin{cases} x = 4 - y \\ z = 1 - 2y \\ w = 2 - y \end{cases}$$

Put $y = 0$, then $(x, y, z, w) = (4, 0, 1, 2)$ is a solution.

Explanation to Theory 1 without equation: A column with a single 1 and 0 elsewhere can help us to replace a material with some compounds.
For example,

	
	0
	1
 + 	0
	0



$$\text{cup of coffee} = 0 \text{ cow} + 1 \text{ coffee bean} + 0(\text{lemon} + \text{leaves}) + 0 \text{ leaf}$$

which means $\text{coffee bean} = \text{cup of coffee}$

With this observation, we can replace **certain materials** by **certain meals**

					
	1	1	0	0	4
	0	2	1	0	1
 + 	0	1	0	1	2
	0	0	0	0	0

					
	1	1	0	0	4
	0	2	1	0	1
	0	1	0	1	2
	0	0	0	0	0

Note that the leaves 🌿 is no longer needed for those packaged materials, we can delete it.

					
	1	1	0	0	4
	0	2	1	0	1
	0	1	0	1	2

which tell us directly the list we want, let's compare the original question

Original Question

Output

	
	4
	1
	2

	
	x
	y
	z
	w

So $x = 4, y = 0, z = 1, w = 2$. ( have not been used.) This process is equivalent as setting $y = 0$ to obtain a solution.

$$\begin{cases} x &= 4 \\ y &= 0 \\ z &= 1 \\ w &= 2 \end{cases}$$

Theory 2: We should stop if

- After deleting zero rows, there are columns that can be rearranged into a triangular matrix with non-zero diagonal.

Example:

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 6 & 6 \\ 2 & 0 & 3 & 2 & 1 & 1 \\ 1 & 0 & 0 & 2 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix} \longrightarrow \begin{pmatrix} 2 & 3 & 1 \\ 0 & 3 & 2 \\ 0 & 0 & 1 \end{pmatrix}$$

We briefly explain why, firstly, circle the entry that corresponding to the diagonal of the triangular matrix

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 6 & 6 \\ 2 & 0 & 3 & 2 & 1 & 1 \\ 1 & 0 & 0 & 2 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

We read equations from bottom to top, one by one

$$x + 0y + 0z + 2w + 1u = 0 \implies x = -2w - 1u.$$

The circled variable would appear as a new variable never appeared before

$$2x + 0y + 3z + 2w + 1u = 1 \implies z = \frac{-2x - 2w - u}{3}.$$

Then the next new variable is y .

$$x + 2y + 3z + 4u + 6w = 6 \implies y = \frac{6 - 6w - 4u - 3z - x}{2}.$$

Therefore, the equation can be solved. We call this process **backwards substituting**.

Identity matrix

Filling the following blanks.

			
?	1	0	0
?	0	1	0
?	0	0	1

	?	?	?
	1	0	0
	0	1	0
	0	0	1

Suppose   , any two can not blend to the third drink (called **linearly independent**). Filling the following blanks.

			
	?	?	?
	?	?	?
	?	?	?

			
	1	0	0
	0	1	0
	0	0	1

This matrix is the **ingradient list of making ingradient**, i.e. do nothing.

Definition 1

The **identity matrix** is a $n \times n$ square matrix with 1 on the diagonal and 0 elsewhere.

Proposition 1

For any $n \times m$ matrix P , $I_n P = P I_m = P$.

Inverse Matrix

The chef is wondering if another guest coming with a special request, so he would like a list to produce the ingredient out of meals.

He has a list

		
	0	2
	1	0

How could he make another list?

		
	?	?
	?	?

He realize this list should have a property, combining them should be.

		
	0	2
	1	0

		
	?	?
	?	?

=

		
	1	0
	0	1

Definition 2

For a $n \times n$ matrix A , an inverse is a matrix B , such that

$$AB = BA = I_n.$$

If such a B exists, A is called **invertible** and denote the inverse as A^{-1} .

Simultaneous Row Operation

For the product $C = AB$, changing raw materials only affect rows of A and C by certain simultaneous row operations. the matrix B would not change since it does not depend on raw materials. Therefore,

Proposition 2

The equality $C = AB$ will still be true if we perform arbitrary simultaneous row operation on A and C

left factor

			
	0	0	2
	0	0	1
	0	2	0
	1	0	0

right factor

		
	2	1
	0	2
	1	1

product

		
	2	2
	1	1
	0	4
	2	1

If A is invertible, we can write $B = A^{-1}C$, this actually tells us that

Corollary 1

If A is invertible, the product $A^{-1}C$ does not change if we perform simultaneous row operation on A and C .

Let me show you an application of this in calculation

$$\begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}^{-1} \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 1 \\ 0 & 2 & 9 \end{pmatrix}$$

$$r_1 \mapsto r_1 - r_3 \quad \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}^{-1} \begin{pmatrix} 1 & 0 & -6 \\ 1 & 2 & 1 \\ 0 & 2 & 9 \end{pmatrix}$$

$$r_2 \leftrightarrow r_3 \quad \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}^{-1} \begin{pmatrix} 1 & 0 & -6 \\ 0 & 2 & 9 \\ 1 & 2 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 0 & -6 \\ 0 & 2 & 9 \\ 1 & 2 & 1 \end{pmatrix}$$

Simultaneous Column Operation

Similarly, Column operations corresponding to updating list when changing final meals by its order, amount, or packing them together. For the product $C = AB$, when final meals changes, the matrices affected is C and B . A is the list of making intermediates, it does not change.

Proposition 3

The equality $C = AB$ will still be true if we perform arbitrary simultaneous column operation on B and C

If B is invertible, we can write $A = CB^{-1}$, this actually tells us that

Corollary 2

If B is invertible, the product CB^{-1} does not change if we perform simultaneous column operation on B and C .

Elementary Matrices

We will introduce elementary matrices.

Let A be an $m \times n$ matrix, and let I_m be $m \times m$ identity matrix. We have a equation

$$A = I_m A$$

After some simultaneous row operation, A changes to A' and I_m changes to E , by what we discussed before, the following equation is true

$$A' = EA$$

We may view this fact by another perspective: The product EA is the same as applying row operations recorded in E to A .

Experiment:

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 2 & 4 \\ 1 & 3 \end{pmatrix} = \begin{pmatrix} 2 & 4 \\ 1 & 3 \end{pmatrix} \implies \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 2 & 4 \\ 1 & 3 \end{pmatrix} = \begin{pmatrix} 3 & 7 \\ 1 & 3 \end{pmatrix}$$

Explanation from another perspective:

left factor	row operation recorded	product: apply row operation
$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$	add 2nd row to 1st row.	$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 2 & 4 \\ 1 & 3 \end{pmatrix} = \begin{pmatrix} 3 & 7 \\ 1 & 3 \end{pmatrix}$

Proposition 4

Suppose E is a matrix obtained by applying some row operations from I_n , then EA is exactly the matrix by applying the same row operations on A .

Definition 3

An elementary matrix E is a matrix after one-step row operation from identity matrix.

Name	Elementary matrices example	Inverse
Switching	$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$	$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$
Multiplying	$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix}$	$\begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & 1 \end{pmatrix}$
Adding	$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$	$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix}$

Exercise: What is the following product? calculate in mind.

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$$

Solution

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 6 & 8 \end{pmatrix} = \begin{pmatrix} 7 & 10 \\ 6 & 8 \end{pmatrix}$$

Look at the following matrix(maybe not elementary). By thinking how it changed from identity matrix, what kind of row operation does it record?

left-factor	row operation recorded.
$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$	Switch two rows
$\begin{pmatrix} 0 \\ 1 \end{pmatrix}$	Delete the first row
$\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$	
$\begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$	
$\begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix}$	

The same logic is also for columns. Recall that $AB = C$ stays as an equation when applying column operations on B and C . **Experiment:**

$$\begin{pmatrix} 2 & 4 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 4 \\ 1 & 3 \end{pmatrix} \implies \begin{pmatrix} 2 & 4 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 6 \\ 1 & 4 \end{pmatrix}$$

Explanation from another perspective:

right factor	row operation recorded	product: apply row operation
$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$	add 1st column to 2nd column.	$\begin{pmatrix} 2 & 4 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 6 \\ 1 & 4 \end{pmatrix}$

Proposition 5

Suppose E is a matrix obtained by applying some column operations from I_n , then AE is exactly the matrix by applying the same column operations on A .

right-factor	column operation recorded.
$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$	Switch two columns
$\begin{pmatrix} 0 \\ 1 \end{pmatrix}$	
$\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$	
$\begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix}$	

What is the meaning of the following product? From the perspective of column action?

$$\underbrace{A}_{3 \times 4 \text{ Matrix}} \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}$$

What is the meaning of the following product? From the perspective of column action?

$$\underbrace{A}_{3 \times 4 \text{ Matrix}} \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

What is the meaning of the following matrix

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \\ 0 & 1 \end{pmatrix} \underbrace{\quad}_A \quad 2 \times 2 \text{Matrix}$$

How about the following

$$\begin{pmatrix} 1 & 0 \\ 1 & 0 \\ 1 & 0 \\ 1 & 0 \end{pmatrix} \underbrace{\quad}_A \quad 2 \times 2 \text{Matrix}$$

Is this product depends on the second row of A ?

Let A be a $m \times n$ matrix, and let I be the 2nd row of A , can you write I in terms of a matrix multiplication?

Let $\vec{v}$ be the 3rd column of A , can you write $\vec{v}$ into a matrix multiplication?